

Nutrition Rules v9 Implementation Audit

Generated August 12, 2026. Updated after re-deriving the Section 7 meal-shape table from the actual ready-meal database and removing coffee allowance behavior.

Result

The v9 checker remains unchanged: each candidate meal must pass calories, scaled protein, scaled fat, and candidate-specific dynamic carbs. The Section 7 table was widened to rounded central database percentiles so the strict checker has more realistic database-matching meal windows.

Area	Status	Evidence
Meal-shape table	Updated	Breakfast 16-30% protein / 25-54% fat; lunch and dinner 18-33% protein / 21-49% fat; snack 13-30% protein / 22-54% fat.
Protein floor	Kept	20% minimum protein floor still applies when building slot profiles.
Carbs	Dynamic	No static meal carb window is stored; candidate carbs are computed from actual candidate protein/fat and the calorie window.
Coffee	Removed	Planner no longer sends coffee fields and backend ignores stale posted coffee fields; no coffee items are added to meals.
Example impact	Improved	For the 90kg/198cm/29 male light lose-weight 5-meal peanut-avoid example, lunch-heavy choices increased from 34 to 44.

Updated Section 7 Table

Meal type	Protein %	Fat %	Carb %
Breakfast	16-30%	25-54%	22-54%
Lunch	18-33%	21-49%	25-54%
Dinner	18-33%	21-49%	25-54%
Snack	13-30%	22-54%	20-54%

Testing Performed

Test command	Result
npm run check	Passed all JavaScript syntax checks.
npm run test:nutrition-v2	Passed 9/9 v9 audit sections with the updated table.
npm run test:nutrition-v9-massive	Passed 29,344 feasible client-plan windows, 5,216 intentional infeasible flags, 102,716 meal slots, and 205,432 boundary candidates.
npm run test:optimizer	Passed ready-meal optimizer matrix: 9/11 complete, 2 explicitly constrained.
npm run test:deterministic-generation	Passed ready-meal generation tests, including stale coffee fields ignored.
npm test	Passed full suite end to end. Only observed runtime note was an existing pg SSL warning, not a test failure.

Operational Notes

The solution was to keep the strict v9 filter but make the fixed Section 7 shape table better match the current meal database. If a meal cannot satisfy the final per-slot windows, it is still rejected instead of being relaxed.